

Rheology of Clarified Fruit Juices. I: Peach Juices

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ABSTRACT

In this work, the rheological behaviour of clarified and depectinated peach juice is studied. Peach juices free of pectin and pulp behave as Newtonian fluids.

The effect of temperature and concentration on the viscosity of these juices is examined. The temperature effect was studied at 10 different temperatures between 5 and 60°C.

Finally, an equation describing the combined effect of temperature and concentration on the viscosity is given.

NOMENCLATURE

A_1	Constant of eqn (4) (dimensionless)
A_2	Constant of eqn (5) ($^{\circ}\text{Brix}^{-1}$)
C	Soluble solids content ($^{\circ}\text{Brix}$)
E_a	Activation energy of flow (kcal mol^{-1} or cal mol^{-1})
K	Constant of eqn (1) ($\text{Pa} \cdot \text{s}^n$)
K_0	Constant of eqn (3) ($\text{Pa} \cdot \text{s}$)
K_1	Constant of eqn (4) $\text{Pa} \cdot \text{s} \cdot ^{\circ}\text{Brix}^{-A_1}$
K_2	Constant of eqn (5) ($\text{Pa} \cdot \text{s}$)
n	Flow behaviour index (dimensionless)

R	Gas constant ($\text{cal mol}^{-1} \text{K}^{-1}$)
T	Temperature (K or $^{\circ}\text{C}$)
τ	Shear stress (Pa)
η	Viscosity (Pa s)
$\dot{\gamma}$	Shear rate (s^{-1})

1 INTRODUCTION

Juices are the most important liquid derivatives of fruits in the food industry. They can be classified according to their pulp content into two main groups: purées and juices, both of which are obtained by crushing or by squeezing (Costell & Durán, 1982a).

Most of the published information is on purées and juices obtained by crushing. These are characterized by a high content of suspended material (Foda & McCollum, 1970; Costell & Durán, 1982b; Costell *et al.*, 1982).

Generally, the rheological behaviour of a clarified juice can be described by a power-law relationship (Vitali *et al.*, 1974; Rao, 1977; Rao & Rizvi, 1986; Ibarz & Pagán, 1987):

$$\tau = K(\dot{\gamma})^n \quad (1)$$

where τ is the shear stress, $\dot{\gamma}$ is the shear rate, K is the consistency coefficient and n is the flow behaviour index.

If the clarified juice was also depectinated, its behaviour would be Newtonian (Saravacos, 1970; Rao & Rizvi, 1986; Ibarz *et al.*, 1987):

$$\tau = \eta(\dot{\gamma}) \quad (2)$$

where η is the coefficient of viscosity.

In this research the rheological behaviour of peach juice was studied. This kind of juice is usually marketed with pulp in suspension. However, in order to obtain the appropriate rheological characteristics in the final product, it was necessary to add depectinated concentrate (by enzymatic elimination).

2 EXPERIMENTAL

2.1 Characterization of the concentrated peach juice

Concentrated juice was manufactured from the most prevalent peach varieties processed in the Lleida region of Spain (Sudanell, Babygold,

Redhaven, Redglove, Springgold). This juice concentrate was industrially depectinated, followed by the removal of suspended particles to obtain a clarified juice. Finally, this juice was concentrated by evaporation.

The soluble solids in the concentrate were determined by refractometry using an Abbe-Zeiss refractometer at a temperature of 20°C. The measured concentration of soluble solids in the concentrated peach juice was 69 °Brix.

A Crison pH-meter was used to measure the pH of the concentrate and gave a value of 3.64 at 20°C. Also, the acidity of the juice concentrate was measured by titration of an 1:11 diluent with NaOH solution using a phenolftaleine indicator. The acidity of the concentrate was 0.64 eq. of acid/litre.

The water activity of the concentrate was measured in a Novasina Humidat-1C 1, and gave a value of 0.775.

Distilled water was used to dilute samples for measurement of rheological behaviour at the different concentrations.

2.2 Rheological measurements

Rheological measurements were carried out on a Rotovisco RV 12 (Haake) concentric cylinder viscometer equipped with a M 500-type measurement attachment which can transmit a maximum torque of 4.90 N cm using an NV-type pair of coaxial cylinders. The viscometer characteristics and the procedure for the measurement have been described elsewhere (Ibarz *et al.*, 1987). A thermostatic bath controls the working temperature within the range 5–60°C ± 1°C.

Rotor speeds were variable in the range 1–512 rpm which enabled rheograms (shear stress versus shear rate diagrams) to be constructed. Readings were taken at increasing rotor speeds until maximum speed was reached, after which it was gradually reduced.

The rheological behaviour of the peach juices at different soluble solids contents (69, 65, 60, 55, 50, 45 and 40 °Brix) and temperatures (5, 10, 15, 20, 25, 35, 45 and 60°C) were studied. All the experiments were replicated.

3 RESULTS

The experimental results of the variation of shear rate ($\dot{\gamma}$) with shear stress (τ) have been fitted to Newton's law (eqn (2)) and to the power-law (eqn (1)). This demonstrated that in all cases the rheological behaviour of the peach juices was Newtonian.

The experimental results (τ , $\dot{\gamma}$) were fitted by the least square method to eqn (2) yielding viscosity values for the different samples. These curve

fitting results are given in Table 1. The fits and the estimates of the viscosity are significant at the 95% probability level. It can be seen from the table that, as expected, the higher the temperature the lower the viscosity, and the higher the soluble solid content the higher the viscosity.

Figure 1 shows the experimental results obtained for the peach juice concentrate at 69 °Brix at different temperatures and the resulting least square lines.

3.1 The effect of temperature

The variation in viscosity with temperature can be described by an Arrhenius-type exponential equation (Saravacos, 1970; Rao *et al.*, 1984):

$$\eta = K_0 \exp\left(\frac{E_a}{RT}\right) \quad (3)$$

where K_0 is a constant, E_a is the activation energy of flow, R is the gas constant, and T is the temperature in K .

TABLE 1
Viscosity of Peach Juices at Different Soluble Solids Content and Temperatures

T (°C)	C (°Brix)	η (Pa s)	r
5	40	0.015 6	0.999 4
	45	0.024 7	0.998 7
	50	0.043 8	0.999 9
	55	0.084 7	0.999 6
	60	0.180 8	0.999 4
	65	0.525 5	0.999 6
	69	2.086 9	0.999 4
10	40	0.012 5	0.998 9
	45	0.019 1	0.999 4
	50	0.033 0	0.999 6
	55	0.063 0	0.999 6
	60	0.123 4	0.999 6
	65	0.404 1	0.999 4
	69	1.288 1	0.999 5
15	40	0.010 7	0.999 3
	45	0.016 1	0.999 1
	50	0.026 9	0.999 4

TABLE 1 — *contd.*

<i>T</i> (°C)	<i>C</i> (°Brix)	η (Pa s)	<i>r</i>
	55	0.047 8	0.999 8
	60	0.092 2	0.999 6
	65	0.249 4	0.999 2
	69	0.743 3	0.999 7
20	40	0.009 0	0.998 9
	45	0.013 2	0.999 2
	50	0.020 9	0.999 6
	55	0.036 8	0.999 8
	60	0.062 3	0.999 7
	65	0.174 8	0.999 2
	69	0.485 3	0.999 0
25	40	0.007 8	0.998 1
	45	0.010 8	0.999 6
	50	0.016 7	0.999 2
	55	0.028 7	0.999 7
	60	0.048 7	0.999 8
	65	0.125 5	0.999 5
	69	0.324 0	0.999 3
35	40	0.006 2	0.997 3
	45	0.008 0	0.998 9
	50	0.011 8	0.999 3
	55	0.019 5	0.999 4
	60	0.032 6	0.999 7
	65	0.070 0	0.999 7
	69	0.155 3	0.999 2
45	40	0.005 0	0.999 0
	45	0.006 0	0.999 3
	50	0.009 0	0.998 8
	55	0.013 2	0.999 1
	60	0.021 4	0.999 5
	65	0.041 3	0.999 7
	69	0.084 2	0.999 5
60	40	0.003 9	0.999 8
	45	0.004 9	0.999 8
	50	0.006 2	0.999 9
	55	0.009 1	0.998 7
	60	0.013 0	0.998 7
	65	0.023 4	0.999 5
	69	0.041 6	0.999 8

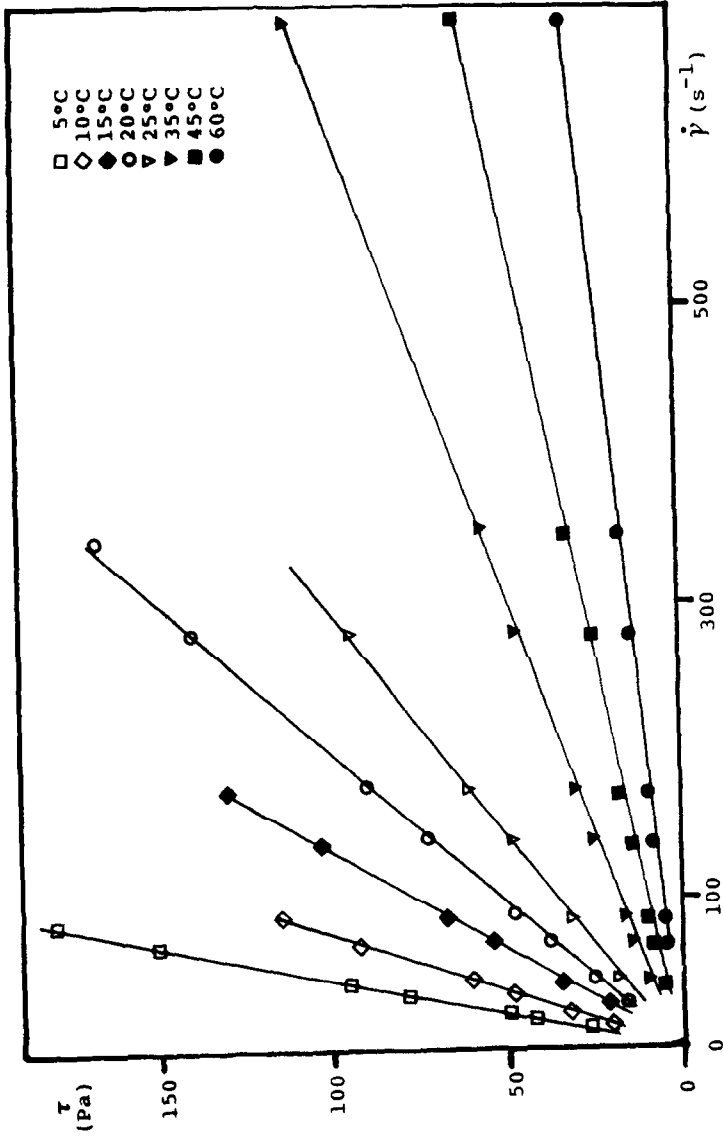


Fig. 1. Rheograms showing the effect of the temperature on juice of 69 °Brix.

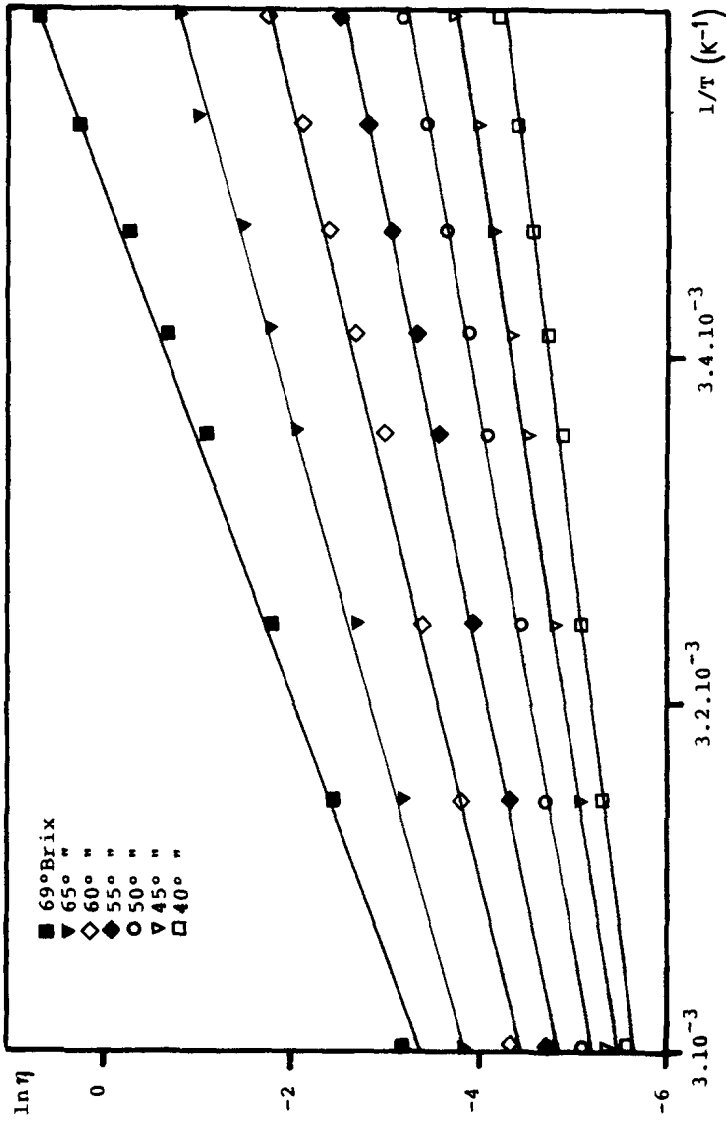


Fig. 2. Change in the viscosity of peach juices with temperature at different soluble solids contents.

In order to obtain the estimates of the parameters of eqn (3), the logarithm of the experimental viscosity was plotted against $1/T$ (Fig. 2).

Table 2 shows the values of the constants K_0 and E_a , and the correlation coefficient, obtained from the fitting of the viscosity data for a given concentration for each temperature. Activation energies for each concentration are obtained from the slope of the corresponding straight line. Both the fits and the estimates of the parameters are significant at the 95% probability level.

From the results, it can be observed that the activation energies of flow increase with the soluble solids content.

3.2 The effect of concentration

Two types of equations describing the variation of viscosity with the soluble solid content can be found in bibliography (Vitali & Rao, 1982; Rao *et al.*, 1984; Ibarz *et al.*, 1987).

For one of these, the viscosity varies with the concentration raised to a given power:

$$\eta = K_1(C)^{A_1} \quad (4)$$

and, for the other the function is exponential:

$$\eta = K_2 \exp(A_2 C) \quad (5)$$

where K_1 , K_2 , A_1 and A_2 are constants and C is the soluble solids content of the juices in °Brix.

The experimental results of viscosity and peach juice concentrations were fitted to the linearized form of eqns (4) and (5) by the least-squared method to obtain the estimates of the parameters of the model. The best fit was found when using the exponential relation (eqn (5)).

TABLE 2
Influence of Temperature on Viscosity. Values of Flow Activation Energy E_a and Constant K_0 for Peach Juices at Different Soluble Solids Content

C (°Brix)	E_a (kcal mol ⁻¹)	K_0 (Pa s)	Corr. coeff.
40	4.63 ± 0.06	3.35 × 10 ⁻⁶	0.995
45	5.54 ± 0.05	9.99 × 10 ⁻⁷	0.992
50	6.52 ± 0.07	3.01 × 10 ⁻⁷	0.995
55	7.55 ± 0.08	8.93 × 10 ⁻⁸	0.996
60	8.73 ± 0.07	2.17 × 10 ⁻⁸	0.995
65	10.71 ± 0.09	1.91 × 10 ⁻⁹	0.997
69	13.29 ± 0.08	7.76 × 10 ⁻¹¹	0.998

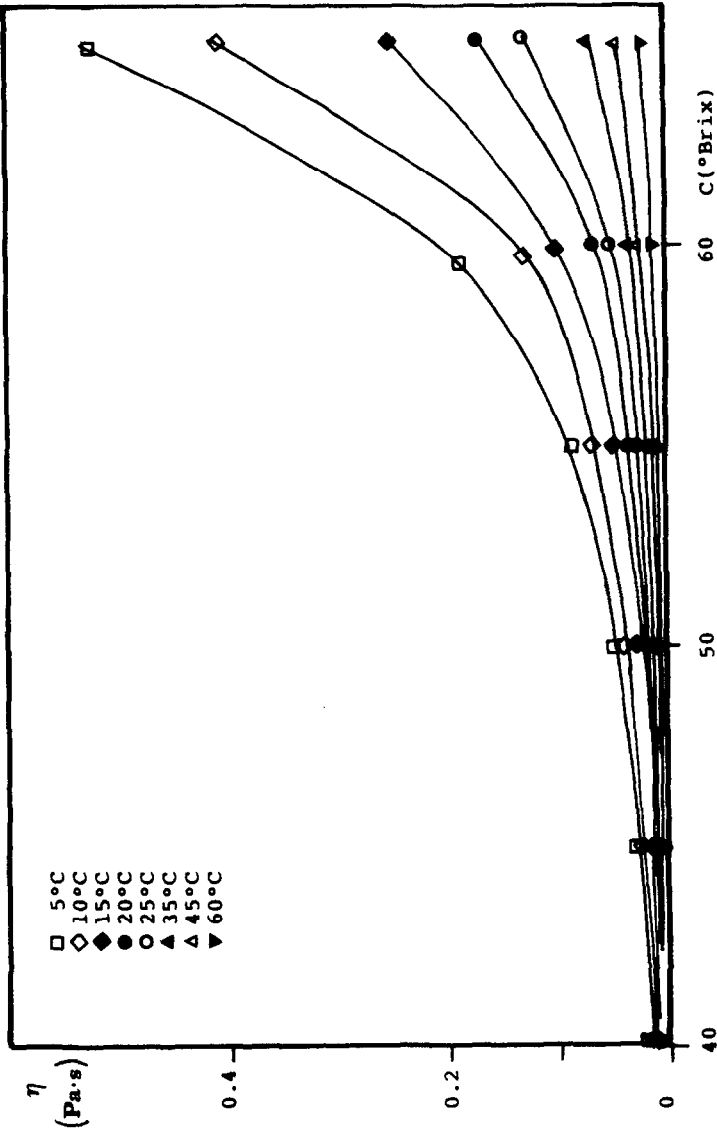


Fig. 3. Change in the viscosity of peach juices with soluble solids content at different temperatures.

Figure 3 shows the experimental results of viscosity versus concentration and the exponential curves obtained.

The estimates of the parameters of the exponential equation at different temperatures and the correlation coefficient are summarized in Table 3. Both the fits and the estimates of the parameters prove to be significant at the 95% probability level.

3.3 The combined effect of temperature and concentration

It would be useful to obtain a single equation describing the combined effect of temperature and concentration on the juice viscosity, or to obtain an equation which relates the parameters that are presented in the Arrhenius equation to the juice concentration.

From Table 2, it can be observed that the activation energy of flow, E_a , and the constant K_0 depend on the soluble solids content. As the juice concentration increases, the activation energy of flow increases and the constant, K_0 , decreases.

The measured values of E_a and K_0 were fitted to different equations including the soluble solids content, resulting in a best fit for the exponential relationship. The equation for E_a was:

$$E_a = 1148 \exp(0.0346 C) \quad (6)$$

with a correlation coefficient, $r=0.997$, and where the concentration and the activation energy of flow are expressed in °Brix, and kcal mol^{-1} respectively. The logarithm of the activation energy of flow versus the concentration in °Brix and the best fit line are shown in Fig. 4.

TABLE 3
Influence of the Soluble Solids Content on Viscosity. Values of the Constants K_2 and A_2 for Peach Juices at Different Temperatures

$T (^{\circ}\text{C})$	$K_2 (\text{Pa s})$	$A_2 (^{\circ}\text{Brix}^{-1})$	Corr. coeff.
5	1.67×10^{-5}	0.161	0.977
10	1.74×10^{-5}	0.155	0.977
15	2.72×10^{-5}	0.141	0.980
20	3.30×10^{-5}	0.131	0.976
25	3.90×10^{-5}	0.125	0.977
35	5.89×10^{-5}	0.109	0.981
45	8.40×10^{-5}	0.096	0.980
60	1.30×10^{-4}	0.080	0.980

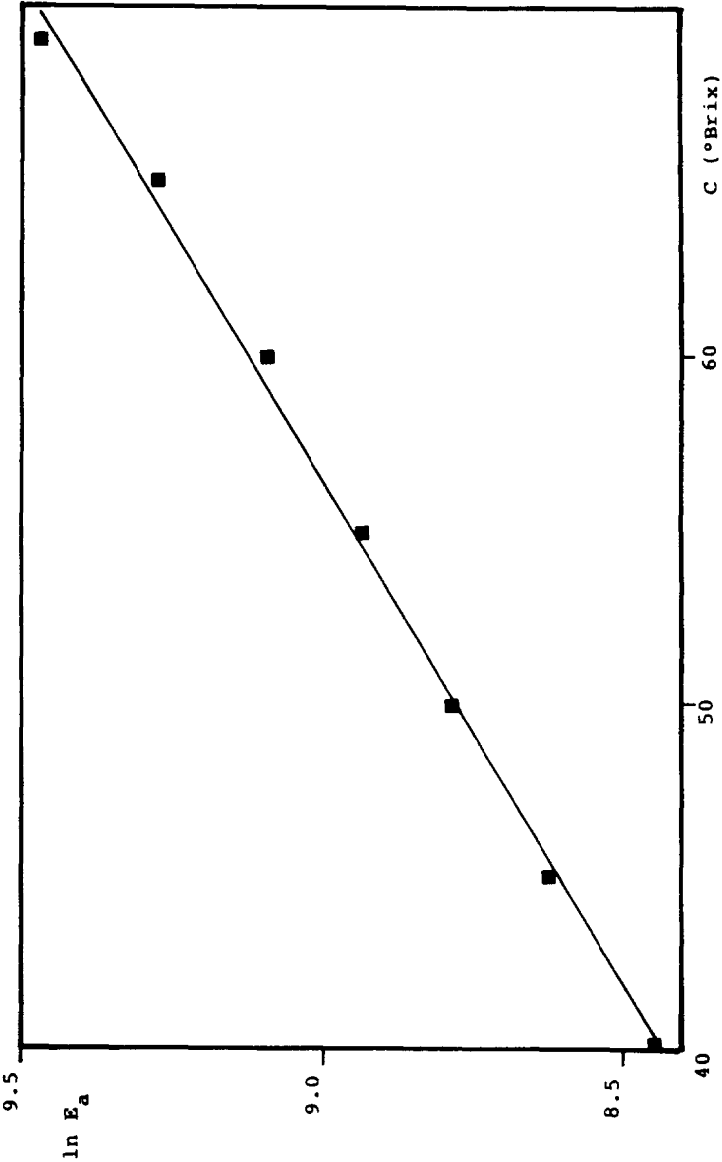


Fig. 4. Change in the flow activation energy E_a with soluble solids content.

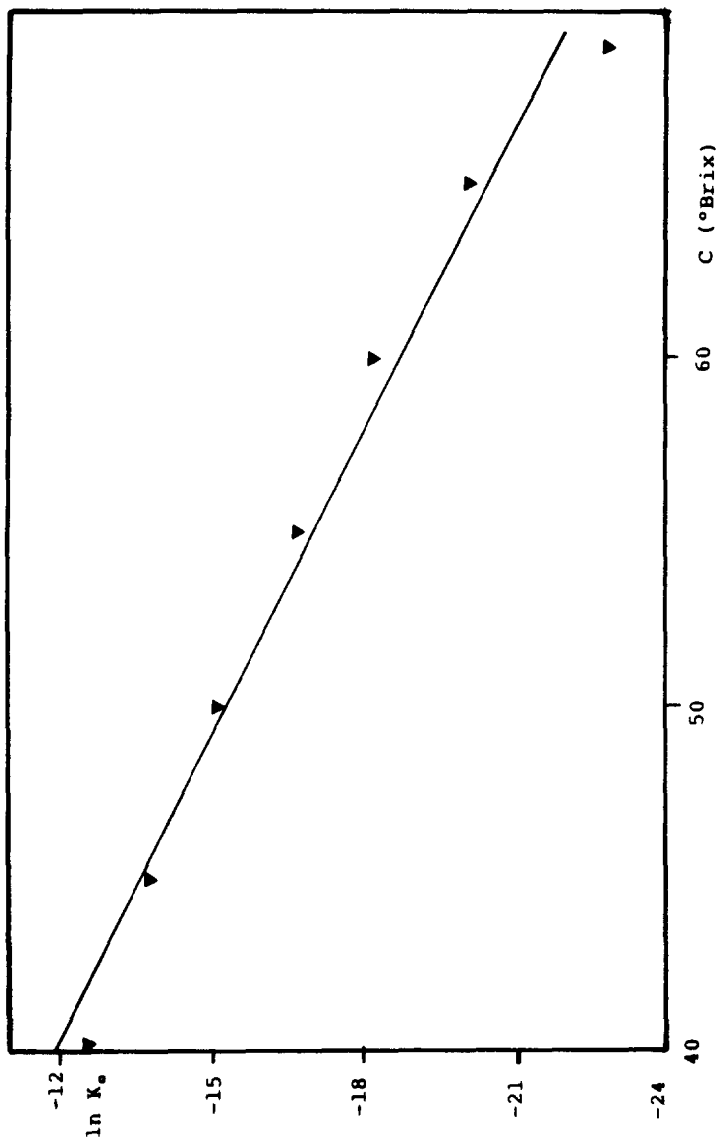


Fig. 5. Change in the Arrhenius constant K_0 with soluble solids content.

The fitted equation for K_0 was:

$$K_0 = 6.46 \exp(-0.343 C) \quad (7)$$

with a correlation coefficient, $r = 0.996$, and where concentration and K_0 were expressed in °Brix and in Pa s respectively. The logarithm of the values of K_0 versus the soluble solids content and the best fit equation are shown in Fig. 5. In these last two equations, both the fit and the estimates of the parameters were significant at the 95% probability level.

Thus, knowing the soluble solids content and the temperature it is possible to estimate the viscosity of any peach juice. Knowing the concentration of soluble solids eqns (6) and (7) are used to calculate the values of E_a and K_0 , and by means of eqn (3) the value of the viscosity is obtained.

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